



Universidad Pontificia Comillas ICAI  
Season 2025/2026



# What is Formula Student?

Formula Student is an **international competition** that brings together the **best engineering schools** and challenges teams made up exclusively of students to **conceive, design, manufacture, and compete** with an electric single-seater built **from scratch**.

In Spain, Formula Student takes place at the iconic **Circuit de Montmeló** in Barcelona and is held every August.

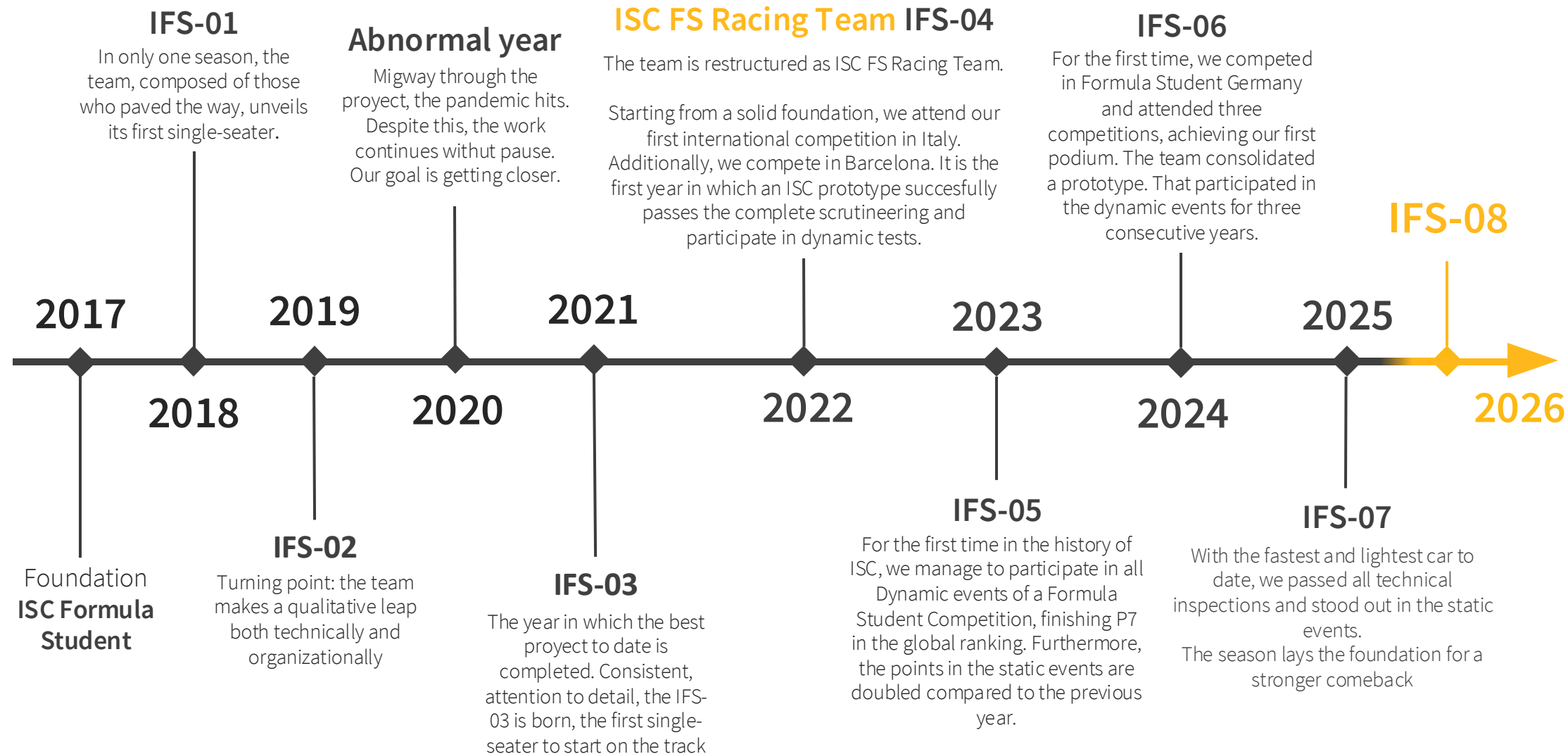
The competition is divided into **two** distinct **parts**: **static events** and **dynamic events**. Between these, teams must **pass a technical inspection** with over a thousand compliance items.

The **static events** focus on engineering **design** justification, **manufacturing cost** analysis, and the development of a **business plan**.

The **dynamic events** are divided into five tests: **acceleration**, **skidpad**, **autocross**, **endurance**, and **efficiency**. These tests aim to push the single-seater to its limits under different scenarios, exploring its top speed, cornering stability, efficiency, and range.



# Our History



# Our evolution

## IFS-01

In just one year, the first team members successfully developed the first prototype in our history: The IFS-01.

With it, they participated in our first competition, Formula Student Spain, in Montmeló. **This single-seater laid the foundation upon which the project would evolve in the following years.**

Competing with the IFS-01 also provided **significant earning and experience** for the team, as they came into **contact with the world of Formula Student for the first time** and had the opportunity to learn from the cars presented by other teams.



## IFS-02

With the first single-seater, the team welcomed more members, which **required an organizational shift**. ISC created **management departments** to better support the Formula Student Team and improve its operations. With a larger and better-organized team, the IFS-02 was developed, showcasing notable technical advancements but still leaving room for improvement.

A new **battery architecture** was introduced, similar to the one used by Tesla in their models, based on cylindrical cells.



# Our evolution

## IFS-03

In the midst of the project, a pandemic disrupted normal operations. However, the team, now composed of over **60 members**, adapted to alternative working methods, allowing the project to move forward with the goal of achieving the first fully functional car.

August 2021 marked a milestone in the team's history with the completion of a fully functional car for the first time. In competition, an outstanding result was achieved in the **Business Plan** event, securing **second place**. Additionally, the excellent feedback received from the judges regarding both mechanical and electrical design provided the team with absolute confidence and enthusiasm heading into 2022.

## IFS-04

Not without challenges, the season was envisioned as an **evolution of the previous prototype**, el IFS03. This strategy allowed the team to focus on development in a way that **paid attention to every detail**, ensuring strong performances in competitions such as Formula ATA Italy 2022 and FS Spain 2022.

By the end of the season, the hard work and effort of the entire year paid off. **For the first time, the team successfully passed scrutineering** (Technical inspections) in both competitions attended during the summer and **participated in Dynamic events at both**.



# Our evolution

## IFS-05

Despite significant challenges at the beginning of the season—including the largest generational turnover the project had ever faced and numerous structural changes to the single-seater, the ISC achieved its best-ever competition results.

After competing in Formula ATA Italy 2023, doubling our points in static events, securing P2 in the business Plan, and climbing three positions compared to 2022, we attended Formula Student Spain. There, we participated in all the dynamic events for the first time in our history, achieving results that marked a turning point for the team: P6 in Business Plan, P6 in Cost Event, P9 in Design Event, DNF in Skidpad, P6 in Acceleration, P9 in Autocross, P6 in Endurance y P5 in Efficiency, achieving P7 in the Overall Ranking.

## IFS-06

The 2023/2024 season leaves us with a sense of accomplishment, marking the **third time we participated in dynamic events** and solidifying our international presence.

We faced significant challenges in the early competitions of the season, such as Formula Student Spain. Additionally, we competed for the first time in **Formula Student Germany**, the flagship competition in the electric category.

The season concluded at Formula Student Italy, where we participated in the **dynamic events and achieved the first-ever podium** in the Business Plan. The IFS-06 overcame harsh weather conditions, **running an endurance race in the rain for the first time**, a mayor achievement for the team given the risks involved.



# Our evolution

## IFS-07: A new beginning

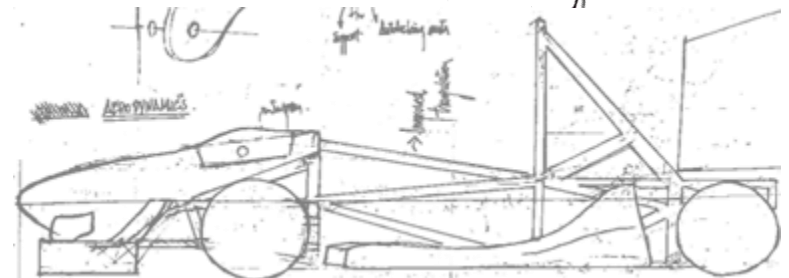
The 2024/2025 season leaves us with the need **to keep pushing**, marking the **fourth time we pass all the technical inspections**.

This year marks a new beginning, bringing again a new prototype, this time with an **aerodynamic package** and an overall optimization of components, which led to the **fastest, and lightest car of our history**.

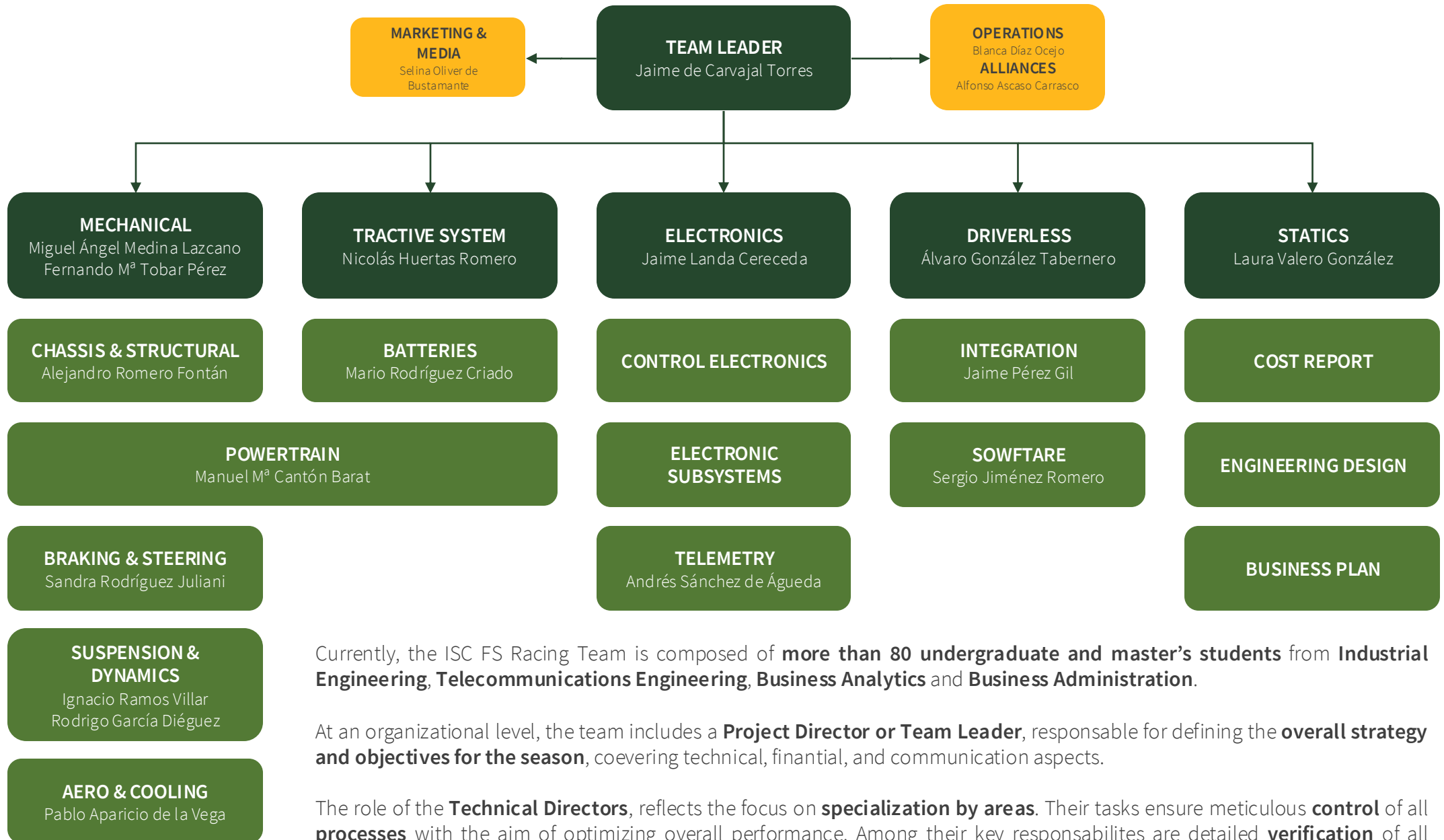
Due to technical difficulties in the endurance, we were not able to finish the competition. However, the team delivered during the static events and throughout the whole competition.

This events made us **stronger** to build a better car for the next season, aiming to **compete in the best competitions** around Europe.

*07 - An advisory to the impossible*



# Our team



Currently, the ISC FS Racing Team is composed of **more than 80 undergraduate and master's students** from **Industrial Engineering, Telecommunications Engineering, Business Analytics** and **Business Administration**.

At an organizational level, the team includes a **Project Director or Team Leader**, responsible for defining the **overall strategy and objectives for the season**, coevering technical, finantial, and communication aspects.

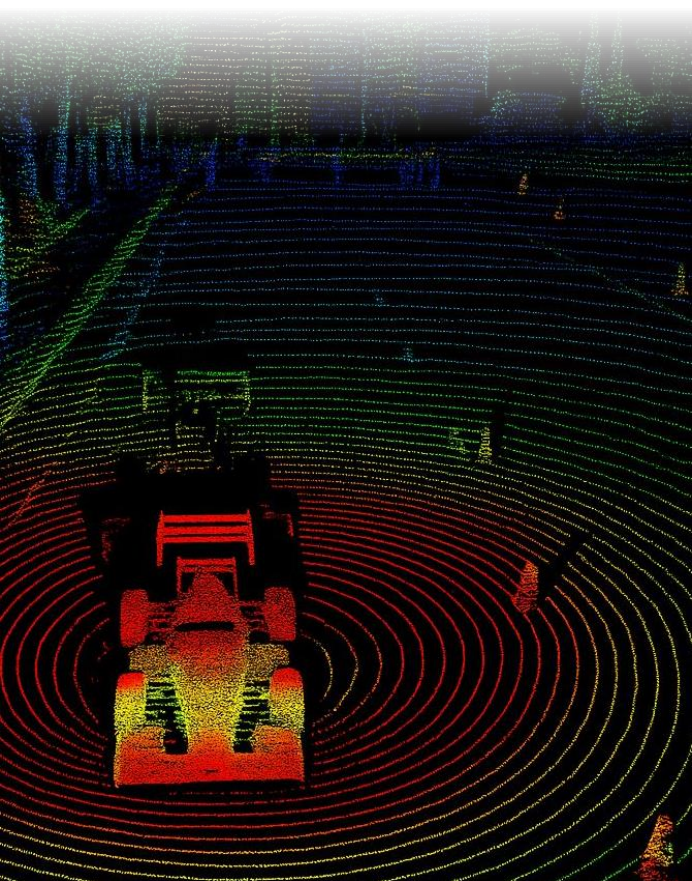
The role of the **Technical Directors**, reflects the focus on **specialization by areas**. Their tasks ensure meticulous **control** of all **processes** with the aim of optimizing overall performance. Among their key responsabilites are detailed **verification** of all **technical aspects** and the **supervision** of each department's **progress**.

# Season 2025/2026

## Promising outlook

The **success in technical inspections** allows for a new perspective within the team for the 2025/2026 season.

| Generational Transition                         | Vehicle concept                                    | Technical Objectives | Autonomous Functionality      |
|-------------------------------------------------|----------------------------------------------------|----------------------|-------------------------------|
| New member selection processes                  | New aerodynamic package                            | Reliability          | Real world implementation     |
| Mentoring program for better knowledge transfer | Dynamic models research                            | Simplification       | Algorithm pipeline validation |
|                                                 | Improvements to the structural part of the vehicle |                      | Further pipeline development  |



# AS at ISC: our stack



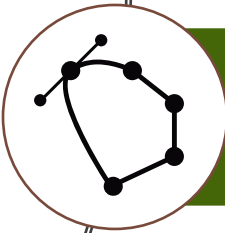
## Perception

New filtering system + terrain understanding we eliminated 75% of false positives.



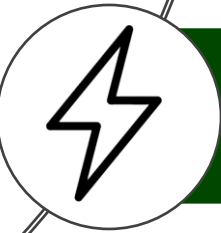
## SLAM

Simultaneous Localization and Mapping, obtained directly from the sensor



## Path Planning

FaSTTUBe algorithm: optimized trajectories through graph search and interpolation.



## Control

PID and Stanley adaptive control: dynamic adjustment of speed and direction, improved stability and lap times

# What do we need in order to grow?

1

## Team

Over 80 committed students, young talent dedicating their free time, passion, and effort.

2

## Experience

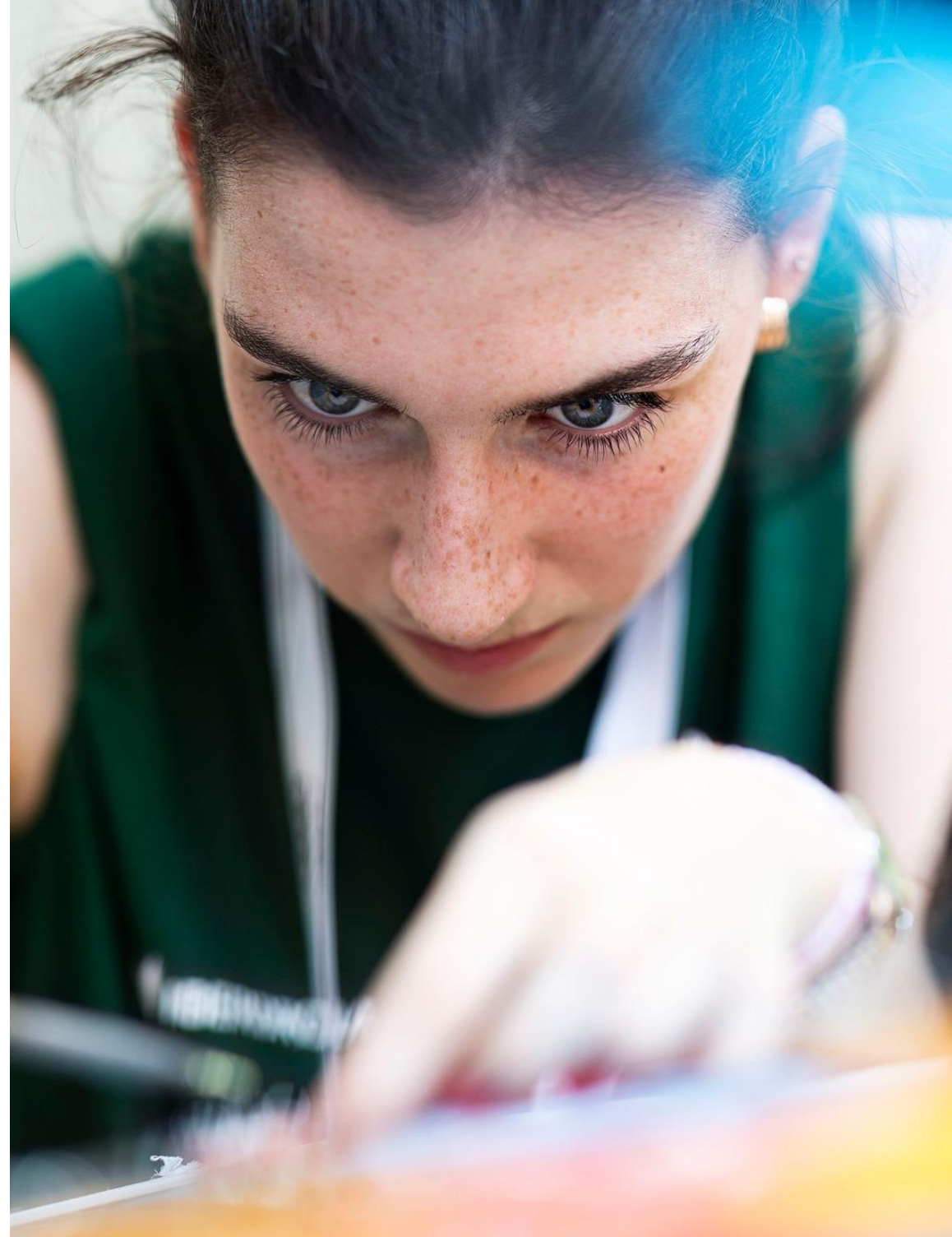
Members ranging from 2nd-year undergraduates to 1st-year master's students. These seven seasons have represented unique learning experiences.

3

## Means

Without our sponsors, nothing would move forward.

With a great team and experience, having more resources is what allows us to keep improving and innovating.



# How to become a partner in this project?

The ISC FS Racing Team looks for in his partners experienced enterprises from various sectors in the industry that share the same **drive for innovation** and see the project as a example of **talent and entrepreneurship** between **young engineers**, which will end their studies soon and land in **work life**.

**Formula Student**, for his **international character**, is the **perfect showroom** for all the companies that admire the potential between the union of engineering and youth.



## LOGISTICS PARTNER

Provides with eans for the team, from vehicle transportation to team member movements, including tools, workshop material or audiovisual support.



## FABRICATION PARTNER

Provide pieces and components for the team or helps with the fabrication of components.



## FINANCE PARTNER

Gives financial aid to the team, supporting specific lines of investigation inside the proyect.

We look for **long-term** sponsorship deals, in which both parts **exploit the potential** of this proyect and its participants to the max.





# What would you be supporting by joining the project?

## Young talent



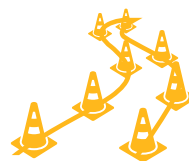
All the members of our team live a unique experience by being part of a similar structure that they will find when they take their first step into the work environment: **reports, deadlines, processes, documentation**; but overall **responsibility**. Overall is a preparation in advance for their next step, while still completing their degrees.

## Innovation in electric mobility

The team carries out an important task in **innovation and formation** in the **electric systems** that are taken on in the mobility industry. Our members acquire the **specific and necessary knowledge** to work on electric mobility ecosystems.



## Innovation in autonomous mobility



We are **leading the first efforts** in implementing **autonomous electric systems** within Formula Student in Madrid, preparing our members with the skills and experience needed to excel in **real autonomous mobility environments**.

## Coaching Spanish engineers

Support to the Spanish engineers and the Spanish universities in the **European and international scene**.



# Our sponsors



NX Technologies



# THANK YOU

Innovation TOGETHER

